

Detecting Hallucinations in Large Language Models via Attention-Head-Level Probes and Voting Ensembles

Anonymous ACL submission

Abstract

Large language models (LLMs) are prone to generating factually incorrect text—so-called *hallucinations*—often accompanied by high confidence. We investigate hallucination detection methods that leverage the model’s internal states without requiring external knowledge retrieval. Using Qwen2-1.5B (28-layer Transformer, 12 heads per layer, hidden size 1536), we conduct a systematic three-stage study. First, we establish baselines comparing perplexity (PPL) and SAPLMA (probe on hidden states), finding that SAPLMA achieves an average AUC of 0.834 versus 0.690 for PPL on a six-topic true/false dataset. Second, we analyze the roles of layer depth and token position, confirming that the middle-upper layers (especially Layer 18) and the last token position carry the richest discriminative signals. Third, we propose a novel per-head probe method: training independent binary classifiers on each of the 336 attention heads’ influence vectors, selecting the top-5 heads by validation AUC, and combining them via a voting ensemble. On a TriviaQA-based test set (200 samples), the summed-heads probe achieves AUC 0.799, surpassing the full SAPLMA baseline (AUC 0.765) with only 5 of 336 heads. The most discriminative heads cluster in layers 11–18, consistent with SAPLMA’s optimal layer range. Our results demonstrate that interpretable, head-level probes can rival or exceed full-hidden-state methods for hallucination detection.

1 Introduction

Large language models have achieved remarkable performance across a wide range of natural language processing tasks. However, a persistent and critical failure mode is their tendency to generate text that appears fluent and confident but contains factual errors—a phenomenon commonly referred to as *hallucination* (Lin et al., 2022; Bai et al., 2022). Hallucination poses significant risks in

high-stakes applications such as medical question answering, legal consultation, and knowledge retrieval, where factual accuracy is paramount.

Existing approaches to hallucination detection can be broadly categorized into two families. *Extrinsic* methods rely on external knowledge bases or retrieval systems to verify generated claims (Manakul et al., 2023), incurring additional latency and infrastructure costs. *Intrinsic* methods, by contrast, exploit the model’s own internal representations—hidden states, attention patterns, or output probabilities—to distinguish truthful from hallucinated outputs (Azaria and Mitchell, 2023; Burns et al., 2022). Intrinsic methods are appealing because they require no external resources and can operate in real time.

In this work, we conduct a systematic investigation of intrinsic hallucination detection methods using Qwen2-1.5B (Team, 2024), a 28-layer Transformer model. Our study proceeds in three stages:

- Baseline Comparison** (Section 3.2): We reproduce and compare two established methods—perplexity-based detection (PPL) and SAPLMA (Azaria and Mitchell, 2023)—on a six-topic true/false dataset, establishing that SAPLMA (average AUC 0.834) significantly outperforms PPL (average AUC 0.690).
- Representation Analysis** (Section 3.3): We investigate how layer depth and token position affect detection quality, confirming an inverted-U performance curve peaking at Layer 18 and demonstrating that the last token position is critical for detection.
- Per-Head Probe Method** (Section 3.4): We propose a novel approach that trains independent binary classifiers on individual attention heads’ influence vectors, selects the top-5 most discriminative heads by validation

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AUC, and combines them via a voting ensemble. This method achieves AUC 0.799 on a TriviaQA test set using only 5 of 336 heads, surpassing the full SAPLMA baseline.

Our key contribution is demonstrating that interpretable, head-level probes can rival or exceed full-hidden-state methods for hallucination detection, while revealing which specific attention heads are most responsible for encoding the model’s “knowledge” of factual correctness.

2 Related Work

Internal Representations of Truth. Azaria and Mitchell (2023) showed that a probe trained on the hidden states of an LLM can predict whether a statement is true or false, achieving significantly better performance than perplexity-based methods. Marks and Tegmark (2023) demonstrated that true/false representations exhibit emergent linear structure in LLM hidden states. Li et al. (2023) proposed inference-time interventions to elicit truthful answers by manipulating internal representations.

Unsupervised Probes. Burns et al. (2022) introduced Contrast-Consistent Search (CCS), an unsupervised method for extracting truth-direction representations from LLM hidden states without labeled data, using contrast pairs and a consistency/informativeness loss.

Attention Head Analysis. Clark et al. (2019) analyzed BERT’s attention patterns to understand what linguistic information different heads capture. Elhage et al. (2022) and Olah et al. (2020) developed frameworks for understanding neural network circuits, including the role of individual attention heads in information processing. Meng et al. (2022) demonstrated that factual knowledge is localized in specific model components, motivating our per-head analysis.

Hallucination Detection. Manakul et al. (2023) proposed SelfCheckGPT for zero-resource hallucination detection. Wei et al. (2024) showed that synthetic data can reduce sycophantic behavior in LLMs. Our work focuses on the intrinsic detection paradigm, extending SAPLMA to the attention-head level.

3 Methodology

3.1 Model and Datasets

Model. We use Qwen2-1.5B (Team, 2024), a causal language model with 28 Transformer layers, 12 attention heads per layer, head dimension 128, and hidden size $d_{\text{model}} = 1536$. The model totals 336 attention heads.

True-False Dataset. Following Azaria and Mitchell (2023), we use a dataset of true/false statements spanning six topics: Cities, Inventions, Chemical Elements, Animals, Companies, and Scientific Facts. Each topic contains paired true and false statements.

TriviaQA Dataset. For Stage 3, we sample questions from TriviaQA (Joshi et al., 2017), generate answers using Qwen2-1.5B with greedy decoding, and auto-label them as correct or hallucinated via fuzzy substring matching against the provided ideal answer aliases.

3.2 Stage 1: Baseline Methods

3.2.1 Perplexity-Based Detection (PPL)

The PPL method is parameter-free: for each statement, we compute the negative log-likelihood (NLL):

$$\text{NLL}(x) = -\frac{1}{T} \sum_{t=1}^T \log p(x_t \mid x_{<t}) \quad (1)$$

where T is the sequence length. Higher NLL indicates the model considers the statement less probable, suggesting a hallucination. An optimal threshold is found by maximizing accuracy on the ROC curve.

3.2.2 SAPLMA: Probe on Hidden States

SAPLMA (Azaria and Mitchell, 2023) extracts the last token’s hidden state from intermediate layers and trains a supervised probe. We extract embeddings from layers 14, 18, 22, 26, and 28 (every 4 layers from the middle to the last layer) and train a three-layer feedforward probe:

$$\text{Probe}(h) = \sigma(W_3 \cdot \text{ReLU}(W_2 \cdot \text{ReLU}(W_1 \cdot h + b_1) + b_2) + b_3) \quad (2)$$

with layer dimensions $256 \rightarrow 128 \rightarrow 64 \rightarrow 1$, ReLU activations, sigmoid output, Adam optimizer, and binary cross-entropy loss. We use leave-one-out cross-validation (5 topics train, 1 topic tests) with 3 random restarts per configuration and 5 training epochs.

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3.3 Stage 2: Representation Analysis

3.3.1 Layer Depth Analysis

We evaluate SAPLMA probes at layers 14, 18, 22, 26, and 28 to characterize how detection quality varies with depth. This reveals the optimal layer for extracting truth-discriminative representations.

3.3.2 Token Position Analysis

At Layer 18 (the optimal layer), we compare probe performance using hidden states from four token positions:

- **First** (position 0): The beginning of the statement.
- **Second** (position 1): The second token.
- **Second-to-Last** (position -2): The penultimate token.
- **Last** (position -1): The final token (default SAPLMA setting).

3.4 Stage 3: Per-Head Probe and Voting

3.4.1 Influence Vector Extraction

In a Transformer, each attention head h at layer l contributes an incremental vector to the residual stream:

$$\Delta_h = O_h \cdot W_O^h \quad (3)$$

where $O_h \in \mathbb{R}^{T \times d_{\text{head}}}$ is the attention output of head h , and $W_O^h \in \mathbb{R}^{d_{\text{model}} \times d_{\text{head}}}$ is the corresponding slice of the output projection matrix. The vector $\Delta_h \in \mathbb{R}^{d_{\text{model}}}$ is the *influence vector* of head h , directly contributing to the final hidden state.

We extract influence vectors for all 336 heads by hooking the input to each layer’s o_proj module, slicing by head, and computing the matrix product with the corresponding weight slice. Features are taken at the answer’s last token position.

3.4.2 Head Selection via Classifier AUC

For each of the 336 attention heads, we train an independent binary classifier (same architecture as the SAPLMA probe) on the training set (Set A, ~ 500 samples). We evaluate each classifier’s AUC on a held-out validation set (Set B, ~ 300 samples) and select the top-5 heads by validation AUC:

$$\text{Top-5} = \arg \text{top}_{h \in \{1, \dots, 336\}} \text{AUC}_h^{\text{val}} \quad (4)$$

Dataset	Accuracy	AUC	Threshold
Cities	0.703	0.740	3.969
Inventions	0.603	0.638	5.233
Elements	0.624	0.662	3.304
Animals	0.624	0.658	4.318
Companies	0.626	0.670	4.041
Facts	0.722	0.775	2.961
Average	0.650	0.690	—

Table 1: PPL-based hallucination detection results across six true/false datasets. The average AUC is 0.690.

3.4.3 Voting Ensemble

On the test set (Set C, 200 samples), we evaluate multiple fusion strategies:

- **Single-Head Probe**: Each of the top-5 heads independently classifies via its own probe.
- **Summed-Heads Probe**: The influence vectors of the top-5 heads are element-wise summed into a single d_{model} -dimensional vector, and one probe is trained on the combined feature.
- **Voting Ensemble**: Each single-head probe outputs a probability; these are binarized via ROC-optimized thresholds, and the final prediction is the majority vote across the 5 probes.

4 Experiments and Results

4.1 Stage 1: PPL vs SAPLMA

Table 1 shows PPL performance across six topics. PPL achieves its best performance on Facts (AUC 0.775) and worst on Inventions (AUC 0.638), with an average AUC of 0.690.

Table 2 presents SAPLMA results across five layers. Layer 18 achieves the best average performance (Accuracy 0.739, AUC 0.834), while the final layer (28) performs worst (AUC 0.761), exhibiting an inverted-U pattern.

Figure 1 visualizes the PPL vs SAPLMA comparison. SAPLMA outperforms PPL on every topic, with an average AUC improvement of 0.144.

4.2 Stage 2: Layer Depth and Token Position

Figure 2 shows how SAPLMA performance varies with layer depth. The left panel confirms the inverted-U pattern, with Layer 18 (64% depth) achieving the best average AUC. The right panel

Layer	Cit.	Inv.	Ele.	Ani.	Com.	Fac.
14	.919	.853	.733	.773	.908	.804
18	.912	.863	.698	.765	.913	.853
22	.903	.775	.774	.775	.878	.834
26	.864	.786	.768	.705	.823	.816
28	.838	.743	.750	.698	.742	.798
Avg	.887	.804	.745	.743	.853	.821

Table 2: SAPLMA AUC by layer and topic. Layer 18 achieves the highest average AUC (0.834). Values shown are average AUC across 3 random restarts with leave-one-out cross-validation.

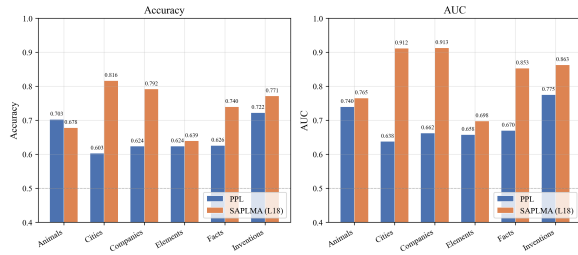


Figure 1: PPL vs SAPLMA (Layer 18) comparison across six topics. SAPLMA consistently outperforms PPL in both Accuracy and AUC.

shows this pattern holds across most topics, though the exact peak varies slightly.

Table 3 and Figure 3 present the token position analysis. The last token dramatically outperforms all other positions (AUC 0.835), while the first and second tokens are near random (AUC ~ 0.50). The second-to-last token shows intermediate performance (AUC 0.668), suggesting a gradual accumulation of semantic information.

4.3 Stage 3: Per-Head Probe and Voting

4.3.1 Head Selection

We trained 336 independent binary classifiers (one per attention head) on Set A and evaluated their AUC on Set B. Figure 4 shows the AUC distribution across all heads. High-performing heads concentrate in layers 11–18, with the top head (L16_H6) achieving AUC 0.731.

The selected top-5 heads are listed in Table 4. All five heads are located in layers 11–18, consistent with SAPLMA’s optimal layer range (Layer 18).

4.3.2 End-to-End Evaluation

Table 5 compares all methods on the TriviaQA test set (200 samples). Figure 5 visualizes the results.

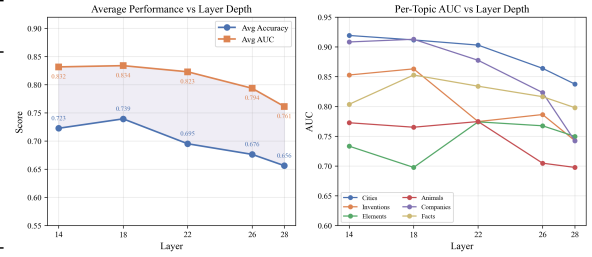


Figure 2: Left: Average Accuracy and AUC across layers. Right: Per-topic AUC across layers. The inverted-U pattern peaks at Layer 18.

Position	Avg Accuracy	Avg AUC
First	0.504	0.500
Second	0.504	0.493
Second-to-Last	0.603	0.668
Last	0.741	0.835

Table 3: Token position analysis at Layer 18. The last token achieves dramatically superior performance.

Key observations:

- The **Summed-Heads probe** (AUC 0.799) surpasses SAPLMA (AUC 0.765) using only 5 of 336 heads’ influence vectors, demonstrating that head-level features can be more discriminative than the full hidden state.
- **Feature-level fusion** (Summed-Heads, AUC 0.799) slightly outperforms **decision-level fusion** (Voting, AUC 0.795), suggesting the head signals are complementary at the feature level.
- **PPL** continues to show weak recall (0.368), struggling to detect hallucinated outputs, though its precision of 0.593 indicates moderate reliability on positive predictions.
- Among single-head probes, **L11_H1** achieves the highest accuracy (0.720) while **L13_H11** achieves the highest single-head AUC (0.763), approaching the full SAPLMA baseline.

5 Analysis and Discussion

5.1 Why Does SAPLMA Outperform PPL?

The performance gap between SAPLMA (AUC 0.834) and PPL (AUC 0.690) can be attributed to four factors:

1. **Information richness:** PPL uses a single scalar (NLL), while SAPLMA leverages a

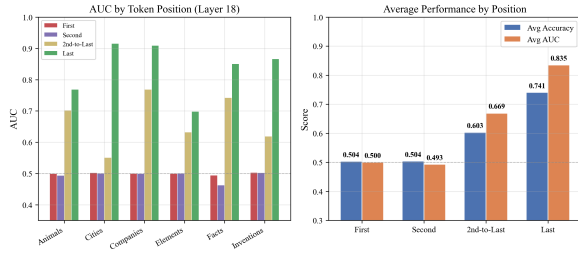


Figure 3: Left: Per-topic AUC by token position. Right: Average performance by position. The last token is critical for hallucination detection.

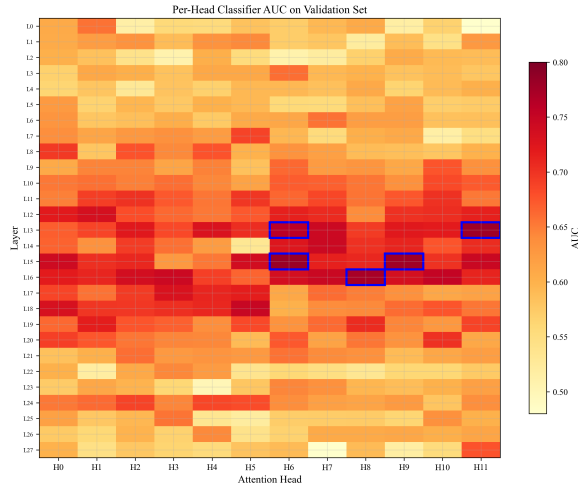


Figure 4: Per-head classifier AUC heatmap (28 layers $\times$ 12 heads). Blue-bordered cells indicate the top-5 selected heads. High AUC values cluster in layers 13–16.

Rank	Head	Layer	Val. AUC
1	L16_H6	16	0.731
2	L15_H6	15	0.729
3	L11_H1	11	0.719
4	L18_H0	18	0.718
5	L13_H11	13	0.717

Table 4: Top-5 attention heads selected by validation AUC. All heads are in layers 11–18.

Method	Acc.	Prec.	Rec.	F1	AUC
PPL	.615	.593	.368	.454	.603
SAPLMA-L18	.700	.667	.621	.643	.765
Single L16_H6	.710	.688	.609	.646	.754
Single L15_H6	.690	.640	.655	.648	.739
Single L11_H1	.720	.707	.609	.654	.746
Single L18_H0	.635	.567	.678	.618	.717
Single L13_H11	.685	.662	.563	.609	.763
Summed-Heads	.735	.681	.736	.707	.799
Voting Ensemble	.725	.722	.598	.654	.795

Table 5: End-to-end hallucination detection on the TriviaQA test set (N=200). The Summed-Heads probe achieves the best AUC (0.799) and Accuracy (0.735).

5.2 The Inverted-U Layer Pattern

The optimal layer for SAPLMA is Layer 18 (64% depth), not the final layer. This inverted-U pattern is consistent with the SAPLMA paper’s findings for other models. A plausible explanation is that middle-upper layers encode the most abstract and task-relevant representations, while later layers begin to transform these into output-oriented representations optimized for next-token prediction, potentially losing some discriminative information.

5.3 Why the Last Token?

The last token’s hidden state (AUC 0.835) vastly outperforms earlier positions (AUC $\sim$ 0.50 for first/second tokens). In autoregressive Transformers, each token’s representation attends to all preceding tokens. The last token has “seen” the entire statement and thus aggregates the fullest semantic representation. Earlier tokens lack this global context.

5.4 Head-Level Interpretability

The concentration of high-performing heads in layers 11–18 (Figure 4) provides interpretable evidence that hallucination-relevant information is localized in specific model components. This is consistent with findings on factual knowledge lo-

1536-dimensional hidden state encoding far richer information.

- Robustness to surface features:** NLL is influenced by token frequency and sentence length—common token sequences may receive low NLL even when factually incorrect. Hidden states encode more abstract semantic representations.
- Nonlinear discrimination:** PPL is inherently a linear threshold classifier, while SAPLMA’s feedforward probe can learn complex nonlinear decision boundaries.
- Internal cognitive signals:** Hidden states directly reflect the model’s internal representation of the input, including signals about whether the model “knows” the statement to be true.

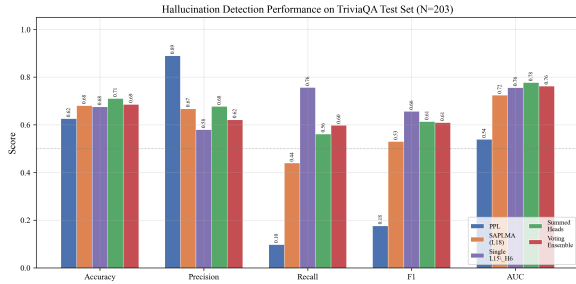


Figure 5: Comparison of all methods on the TriviaQA test set across five metrics. The Summed-Heads probe achieves the best overall performance.

calization (Meng et al., 2022) and circuit analysis (Olah et al., 2020). The fact that only 5 of 336 heads suffice to surpass SAPLMA suggests that truth-discriminative signals are highly concentrated rather than uniformly distributed.

5.5 Feature-Level vs Decision-Level Fusion

The slight advantage of Summed-Heads (AUC 0.799) over Voting (AUC 0.795) suggests that the top-5 heads capture complementary aspects of the truth/hallucination distinction, and that combining their signals at the feature level allows the probe to learn richer joint patterns than majority voting at the decision level.

6 Limitations

Our study has several limitations. First, the true/false dataset uses a constrained statement format that may not fully represent the diversity of real-world hallucinations. Second, the TriviaQA-based evaluation relies on fuzzy string matching for auto-labeling, which introduces annotation noise. Third, we evaluate on a single model (Qwen2-1.5B); generalization to larger models and different architectures remains to be verified. Fourth, the per-head probe method requires a forward pass through the model for feature extraction, which adds computational overhead compared to PPL. Finally, our voting ensemble uses a simple majority rule; more sophisticated fusion strategies may yield further improvements.

7 Conclusion

We conducted a systematic three-stage study of hallucination detection using the internal states of Qwen2-1.5B. Our key findings are:

1. SAPLMA (AUC 0.834) significantly outperforms PPL (AUC 0.690), confirming that hid-

den states contain richer truth-discriminative information than output probabilities.

2. The optimal detection layer is Layer 18 (64% depth), not the final layer, and the last token position is critical for effective detection.

3. A novel per-head probe method, using only 5 of 336 attention heads’ influence vectors, achieves AUC 0.799 on a TriviaQA test set—surpassing the full SAPLMA baseline (AUC 0.765) with 300× fewer features.

4. The most discriminative heads cluster in layers 11–18, providing interpretable evidence for the localization of hallucination-related signals in Transformer models.

These results demonstrate that interpretable, head-level probes can rival or exceed full-hidden-state methods for hallucination detection, opening new directions for efficient and explainable detection systems.

Acknowledgments

We thank the anonymous reviewers for their valuable feedback. This work was completed as a course project for Natural Language Processing.

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